

Food Borne Diseases — Part 1: Foundations + Enteric Fever

Compact A3 landscape study sheet

Definition + Main Classification

Food borne diseases: illness caused by consumption of food/water contaminated with **bacteria and/or toxins, parasites, viruses, or chemicals**.

- **Food-borne infections:** caused by consuming foods contaminated with bacteria, viruses, or parasites.
- **Food-borne intoxications:** caused by consuming foods already contaminated with a toxin.

Sources of toxins: certain bacteria (pre-formed toxins), poisonous chemicals, natural toxins in animals/plants/fungi.

Examples + Subclassification + Causes of Intoxication

Examples of food-borne infections

- **Bacterial:** Typhoid/paratyphoid fever, brucellosis, cholera.
- **Viral:** Viral hepatitis A & E, poliomyelitis.
- **Parasitic:** Amoebiasis, giardiasis.

Food-borne infections may be:

- **Enteric:** mainly GI symptoms (e.g. *Salmonella enteritidis*, *E. coli* O157:H7, *Campylobacter jejuni*, *Vibrio cholerae*, Rotavirus).
- **Systemic:** start in GIT then spread by blood (e.g. typhoid fever, brucellosis).

Causes of intoxications: bacterial toxins (*C. botulinum*, *C. perfringens*), aflatoxins, poisonous mushrooms, ciguatera fish poisoning, chemical toxins (insecticides), heavy metals (lead, mercury).

Food-borne Infection vs Intoxication

Feature	Food-borne infections	Food-borne intoxications
Cause	Bacteria / viruses / parasites	Toxin
Mechanism	Invade and/or multiply in intestinal lining	No invasion or multiplication
Incubation	Hours to days	Minutes to hours
Symptoms	Diarrhea, nausea, vomiting, abdominal cramps ± fever	Vomiting, nausea, diarrhea; double vision, weakness, respiratory failure, numbness, sensory & motor dysfunction
Transmission	Can spread person-to-person via feco-oral route	Non-communicable
Food contamination factors	Inadequate cooking, cross contamination, poor personal hygiene, bare-hand contact	Inadequate cooking, improper holding temperature

Enteric Fever — Overview + Epidemiology + Source + Transmission

Enteric fever includes **typhoid** and **paratyphoid**; acute **systemic** infections with fever.

Causative organisms: *Salmonella typhi* and *Salmonella Paratyphi* A, B, C. They remain viable in water, milk, ice-cream and other foods for weeks, but are destroyed by heat and disinfectants.

Incidence & distribution

- **Age:** more prevalent in **10–30 years**.
- **Sex:** males affected more; females more often carriers.
- **Season:** all year, more in summer.
- **Socio-economic:** common in low-/middle-income settings with unsafe water, poor sanitation, poor hygiene.

Source of infection: only **human** — cases and carriers (fecal/urinary). Types of carriers: incubatory, convalescent, contact/healthy, chronic.

Mode of infection: feco-oral via contaminated food/water; hand-to-mouth from case/carrier; fomites using soiled articles.

Pathogenesis + Clinical Picture + Communicability + Case Definition

Pathogenesis: once ingested, salmonella survive stomach, enter small intestine, penetrate intestinal wall, are engulfed by macrophages, then spread through lymphatics/blood to RES causing systemic disease.

Clinical picture:

- **1st week:** gradual **step-ladder fever (38–40°C)**, headache, dry cough, fatigue, malaise, anorexia, abdominal discomfort/constipation.
- **2nd week:** sustained high fever, toxic look, abdominal distension, splenomegaly, coated tongue, rose spots may appear.
- **3rd week:** complications may occur (intestinal hemorrhage/perforation, delirium, severe toxemia).
- **4th week:** gradual improvement in uncomplicated cases.

Period of communicability: infectious during last days of incubation, throughout illness, and for varied periods in convalescents or permanent carriers.

Suspected case: prolonged fever (≥3 days) compatible with typhoid, no other apparent source. **Probable case:** suspected case plus epidemiologic link / suggestive lab evidence. **Confirmed case:** laboratory confirmation of *S. typhi/paratyphi*.

Prevention

General prevention

1. Environmental sanitation: safe water, sanitary waste disposal, fly control, food sanitation.
2. Health education for clean habits, especially hand washing after defecation/urination.
3. Special sessions for food handlers about mode of infection and methods of prevention.

Typhoid Fever Vaccines — updated

Vaccine	Approved age	Dose & route	No. doses	Booster	Efficacy
Vi capsular polysaccharide vaccine (ViCPS) — Typhim Vi	≥2 years	0.5 mL IM injection	1	Every 2 years	80%
Live attenuated Ty21a vaccine — Vivotif	≥6 years	1 capsule orally every other day	4	Every 5 years	60%

Control

Domain	Measures
Cases	Notification; isolation at home if sanitary requirements fulfilled otherwise fever hospital; concurrent and terminal disinfection; appropriate antibiotics according to local resistance (e.g. ceftriaxone, azithromycin, or fluoroquinolones if sensitive); release when afebrile for 48–72 h, symptoms resolved, full antibiotic course completed, and at least one negative stool/urine culture 48 h after antibiotics. High-risk food handlers may require 2–3 consecutive negative cultures 24–48 h apart.
Contacts	Enlist contacts and keep under surveillance for 3 weeks from last exposure; active immunization; search for carriers among contacts.
Environment	Super-chlorination of water supply; closure of restaurants not meeting sanitary standards; fly control; exclude typhoid carriers from food handling.
Epidemic measures	Super-chlorination of water, sanitary waste disposal, strict food and milk sanitation; public health education; vaccination of at-risk population; epidemiologic tracing of infection source.

Brucellosis — Overview, Sources, Transmission, Clinical Picture

Brucellosis: zoonotic bacterial infection caused by genus *Brucella*; transmitted from animals to humans; may be acute or chronic.

Causative organisms: *B. melitensis* (goats/sheep; most pathogenic), *B. abortus* (cattle), *B. suis* (pigs), *B. canis* (dogs).

Source of infection: only animals — infected cattle, goats, sheep, pigs. Organisms occur in excreta, discharge, tissues, mammary glands, placenta, and vaginal discharge of aborted animals.

Mode of transmission:

- Food-borne: infected milk or dairy products.
- Contact with infected animals through wounds.
- Airborne: inhalation of aerosolized secretions.

Occupational disease: farmers, veterinarians, abattoir workers, laboratory workers.

Incubation period: **1–3 weeks**.

Clinical picture

- **Acute phase:** fever (undulant/intermittent), night sweats, malaise, headache, muscle aches, weight loss.
- **Localized disease:** osteoarticular involvement, hepatosplenomegaly, other focal complications.

Diagnosis: chronic vague picture should raise suspicion; **blood culture is diagnostic**.

Brucellosis — Prevention, Occupational Safety, Control

Prevention: animal measures: sanitary clean environment, veterinary care, sanitary management of aborted animals, vaccination of young calves in endemic areas, agglutinin survey for animals.

Prevention: human measures: milk sanitation by boiling/pasteurization, meat sanitation and proper cooking, health education, **no approved human vaccine**.

Occupational safety

- Use gloves, masks, protective clothing for veterinarians, farmers, abattoir and laboratory personnel.
- Frequent handwashing and disinfection of equipment after handling animals or animal products.
- Training and awareness for high-risk workers.

Control

- **Cases:** notification; no strict control needed because no person-to-person transmission; treatment.
- **Source investigation:** test suspected animals, segregate reactors, slaughter diseased animals, stop suspicious milk supply until control is ensured.

Cholera — Overview, Agent, Reservoir, Pathogenesis, Clinical Picture

Cholera is caused by *Vibrio cholerae* and is usually transmitted by contaminated water or food. It affects the intestines and may lead to severe dehydration, watery diarrhea, and death if untreated.

Causative agent: serogroups **O1** and **O139** produce cholera toxin (CT). O1 has biotypes **Classical** and **El Tor**; O139 causes localized outbreaks.

Reservoir: only human — case or carrier. **Mode of infection:** ingestion of contaminated food/drinks. **Incubation:** few hours to **5 days** (usually 2–3 days).

Pathogenesis: ingestion → colonization of small intestine → toxin production → massive water/electrolyte secretion → watery diarrhea.

Clinical picture:

- Acute watery diarrhea; hallmark **rice-water stools**.
- Vomiting.
- Severe dehydration: dry skin, oliguria, hypotension, shock.
- Muscle cramps from electrolyte imbalance.
- Severe cases may progress rapidly to hypovolemic shock and death.

Diagnosis: stool examination or rectal swab.

Cholera — Prevention + Control

General prevention: sanitary clean environment; public health education for clean habits; international quarantine measures for travelers/pilgrims from endemic/infected areas.

Specific prevention:

1. Parenteral whole-cell vaccine has little practical value: about **50% protection for 3–6 months**.
2. New oral vaccines (killed whole-cell or attenuated) give higher protection for about **3 years**.
3. Used for residents/travelers to endemic areas and pilgrims.
4. **Chemoprophylaxis (preferred):** Tetracycline **0.5 g every 6 h for 3 days** for contacts, travelers to/from endemic or infected areas, and returning pilgrims.

Control of cases: notification; hospital isolation only in severe cases; treatment with fluid therapy (for dehydration) and tetracycline; concurrent and terminal disinfection.

Control of contacts: surveillance of people who shared food/drink for **5 days** from last exposure; chemoprophylaxis (adults tetracycline 500 mg QID; children 50 mg/kg/day in 4 divided doses); investigate source such as polluted water or contaminated food.

Five Keys to Safer Food (WHO-style safety reminders)

1) Keep clean

- Wash hands before handling food and often during preparation.
- Wash hands after going to the toilet.
- Wash and sanitize surfaces and equipment used for food preparation.
- Protect kitchen areas and food from insects, pests and other animals.

Why? Microorganisms are widely found in soil, water, animals and people; hands, cloths and utensils can transfer them to food.

3) Cook thoroughly

- Cook food thoroughly, especially meat, poultry, eggs and seafood.
- Bring soups and stews to boiling; aim for **70°C**.
- For meat/poultry, make sure juices are clear.
- Reheat cooked food thoroughly.

Why? Proper cooking kills almost all dangerous microorganisms.

5) Use safe water and raw materials

- Use safe water or treat it to make it safe.
- Select fresh and wholesome foods.
- Choose processed foods such as pasteurized milk.
- Wash fruits and vegetables, especially if eaten raw.
- Do not use food beyond its expiry date.

Why? Water and raw materials may be contaminated with microorganisms or chemicals; damaged/mouldy foods may contain toxins.

2) Separate raw and cooked

- Separate raw meat, poultry and seafood from other foods.
- Use separate equipment and utensils (knives, cutting boards) for raw foods.
- Store food in containers to avoid contact between raw and prepared foods.

Why? Raw foods may contain dangerous microorganisms that can cross-contaminate cooked/ready foods.

4) Keep food at safe temperatures

- Do not leave cooked food at room temperature for more than **2 hours**.
- Refrigerate cooked and perishable food promptly (preferably below **5°C**).
- Keep cooked food piping hot (more than **60°C**) before serving.
- Do not store food too long even in the refrigerator; do not thaw frozen food at room temperature.

Why? Microorganisms multiply quickly at room temperature.

Definition, Virus Types, Reservoir, Transmission

Poliomyelitis: acute infectious viral disease caused by **poliovirus**; mainly affects young children and may cause paralysis.

Serologic types: **Type I** present in epidemic and isolated from paralytic cases; **Type II** present in endemic disease; **Type III** less frequently isolated.

Reservoir: Humans are the primary reservoir. Carriers may be asymptomatic or symptomatic. Contaminated water/food may temporarily act as **secondary environmental reservoirs**. **No animal reservoir**.

Transmission: dominant **feco-oral route**; respiratory transmission less efficient but can occur in close-contact settings. In developing countries both may occur, but feco-oral is most important.

Paralytic Polio Types + Fatality + Clinical Picture + Diagnosis

Paralytic polio types:

- **Spinal polio 79%:** asymmetric paralysis, legs affected most often.
- **Bulbar polio 19%:** weakness of facial, oropharyngeal, and respiratory muscles supplied by cranial nerves.
- **Bulbospinal polio 2%:** combination of bulbar and spinal paralysis.

Fatality: **2–10%** due to respiratory muscle paralysis.

Post-polio syndrome: after a latent period of **15–20 years** after paralysis, muscle pain may worsen with more weakness/paralysis; more common in females.

Clinical picture:

- **Inapparent / abortive:** mild flu-like symptoms, diarrhea, fever, sore throat, malaise.
- **Non-paralytic:** flu-like symptoms with neck/back stiffness, sometimes nausea, vomiting, abdominal pain.
- **Paralytic:** starts 7–10 days after onset; muscles depend on type; paralysis is **asymmetric** and **flaccid**; legs more affected than arms; risk is increased by IM injection during incubation, tonsillectomy, and excessive muscular activity.

Diagnosis: clinical picture; demonstration of virus in stool, CSF, or oropharyngeal secretions; virus antibodies in serum.

Pathogenesis + Outcome of Infection

Pathogenesis

- Virus enters through the mouth by feco-oral route.
- Replication in **oropharynx** and **gastrointestinal tract**.
- Invades local lymphoid tissue, may enter bloodstream, then infect CNS cells.
- Destruction of motor neurons leads to characteristic paralysis.
- **Incubation: 7–14 days**.

Outcome of infection:

- **Asymptomatic: 70%**.
- **Abortive polio:** about **24%**.
- **Non-paralytic polio:** aseptic meningitis in **1–5%**; symptoms last 2–10 days, then complete recovery.
- **Paralytic polio: <1%** result in flaccid paralysis.

Communicability + Distribution + Prevention + Control

Period of communicability: highest infectiousness just before and after symptom onset; virus may remain in stool up to **6 weeks**. Susceptibility is common but paralysis is rare; type-specific long immunity follows apparent or inapparent infection.

Distribution: mainly children **<5 years**; disease of low social class; endemic in bad sanitary environment, whereas with good sanitation it appears mainly as epidemics affecting older groups.

Prevention: environmental sanitation; health education; vaccination (oral or inactivated polio vaccines).

Routine immunization in Egypt: **7 doses** at birth, 2nd month, 4th month, 6th month, 9th month, 1 year, and 1.5 years. **National campaigns:** all children aged 1–5 years even if already recently vaccinated.

Control:

- **Cases:** notification, isolation, symptomatic treatment.
- **Contacts:** surveillance for **14 days**; avoid predisposing factors that increase risk of paralysis.

Eradication: high immunization coverage (≥3 OPV doses), AFP surveillance with lab confirmation, nationwide mass campaigns (two OPV doses one month apart for children under 5), “mopping up” vaccination in high-risk areas, and vaccine cold-chain quality assurance.

Sabin (OPV) vs Salk (IPV)

Feature	Sabin (OPV)	Salk (IPV)
Vaccine type	Live attenuated poliovirus	Inactivated (killed) poliovirus
Administration	Oral	Injection
Immunity	Local and systemic	Systemic
Cost	Lower cost	More expensive
Safety	Rare risk of vaccine-associated paralytic polio (VAPP)	No risk of VAPP
Use in eradication	Widely used in eradication campaigns	Preferred in polio-free / low-risk areas
Herd immunity	Promotes herd immunity through virus excretion	Does not contribute to herd immunity (individual immunity)
Main concern	Vaccine-derived polio outbreaks (VDPV)	Less effective in stopping transmission

Polio in Egypt + Worldwide Endemic Areas

Polio situation in Egypt

- Egypt was declared **polio-free in 2006** after eliminating wild poliovirus through extensive immunization campaigns; **last reported case was in 2004**.
- In 2021, Egypt experienced an outbreak of **circulating vaccine-derived poliovirus type 2 (cVDPV2)**, first detected in 2020, but no paralytic polio cases were reported.
- Egypt maintains strong **AFP surveillance** and environmental monitoring for early detection and response.

Worldwide endemic areas of polio

- **2005:** Pakistan, Afghanistan, Egypt, India, Niger.
- **2022 wild-virus endemic countries:** Pakistan and Afghanistan (according to WHO).

Hepatitis A — Definition, Epidemiology, Transmission, Mechanism

Hepatitis A (HAV) is a highly contagious liver infection caused by a **non-enveloped RNA virus** of family **Picornaviridae**. It is transmitted by the **fecal-oral route** and may cause epidemics, especially via contaminated water.

Infections:

- **Developing countries:** infection usually occurs early in life with few or no symptoms.
- **Developed countries:** often causes acute viral hepatitis in adolescents and adults.
- **No chronic infection.**

Transmission:

1. Ingestion of contaminated food/water; waterborne outbreaks are associated with sewage-contaminated or inadequately treated water. **Shellfish** are particularly associated.
2. Direct contact with an infectious person.

Mechanism: HAV enters via contaminated food/water/contact → reaches liver through bloodstream → infects hepatocytes → viral replication triggers immune response → cytotoxic T cells attack infected hepatocytes causing inflammation/jaundice/fatigue → infection is self-limited and lifelong immunity develops.

Hepatitis A — Prevention + Vaccines + Schedule

General prevention

1. Environmental sanitation: safe drinking water and proper sewage disposal.
2. Personal hygiene: frequent handwashing with soap after toilet use, before preparing food, and after caring for ill people.
3. Food safety: avoid raw/undercooked shellfish or unhygienic foods; cook food thoroughly; wash fruits and vegetables with clean water.
4. Food handlers should wash hands properly and maintain hygiene.

Specific prevention: HAV vaccination

- **Inactivated vaccines** (most common): e.g. **Havrix**, **Vaqta**.
- **Combined vaccine:** **Twinrix** (Hep A + B).
- **Live attenuated vaccines:** used in some countries such as China.

Schedule: first dose at any time, typically from **1 year of age**; second dose **6–18 months later** (booster). Gives long-term, potentially lifelong immunity.

Hepatitis A — Symptoms, Outcome, Diagnosis

Incubation period: about **2–6 weeks** (average **28 days**).

Symptoms range from mild to severe.

Prodromal phase: fatigue, nausea/vomiting, abdominal pain (especially right upper quadrant), fever, loss of appetite.

Icteric phase: jaundice, dark urine, pale stool, itching. Some cases are asymptomatic, especially in children.

Outcome: almost everyone recovers fully with **lifelong immunity**; a very small proportion may die from **fulminant hepatitis**.

Diagnosis

- **Anti-HAV IgM:** recent/acute infection; detected at symptom onset, peaks at 4–6 weeks, persists 3–12 months.
- **Anti-HAV IgG:** past infection or vaccination; provides immunity.
- Elevated liver enzymes (ALT, AST).
- Bilirubin may be elevated in jaundiced patients.
- **Epidemiologic links:** travel to endemic areas, contaminated food/water, or contact with infected individual.

Hepatitis A — Indications, Effectiveness, Passive Immunoprophylaxis

Indications for vaccination

- Routine vaccination for children aged **1 year or older** in many countries.
- **High-risk groups:** travelers to high/intermediate endemic areas (4 weeks before departure), people with chronic liver disease, homosexuals, drug abusers, laboratory workers handling HAV, and during outbreaks.

Effectiveness: protective antibodies develop in **>95%** after first dose and nearly **100%** after second dose. Immunity lasts at least **20 years** and may be lifelong.

Passive immunoprophylaxis (immune globulin, IG): human IgG preparation giving short-term protection or post-exposure prophylaxis. It should be given **within 2 weeks of exposure**.

Indications for IG: household/close contacts of cases, foodborne or waterborne outbreak exposure.

Post-exposure dose: **0.02 mL/kg single dose**.

Hepatitis E — Key Summary

Feature	Summary
Transmission	Primarily fecal-oral , often via contaminated water or food. Waterborne spread is common where sanitation is poor. Zoonotic transmission may occur from animals such as pigs, deer, and goats.
Incubation period	2–9 weeks (average 6 weeks).
Severity	Can be severe, especially in pregnant women and immunocompromised individuals.
Chronicity	Rarely chronic, but can become chronic in immunocompromised people.
Vaccine	Available in China , not globally widespread.
Prevention	Good sanitation and hygiene, safe water, avoid undercooked meat, vaccination in China.